

CAST / Causal educator kit

One person, two possible paths

Causal effects across time and people

60–90 minute class · Synthetic examples

Editable draft with a study guide, R lab and demo data

The missing comparison

Hector receives treatment and improves.

What would have happened to *Hector*, at the *same time*, without treatment?

One assigned path can be observed.

The alternative remains unobserved.

Before versus after compares two times on one path.

Treatment versus control compares two potential paths at one time.

A survival snapshot

At horizon t , let $A^w(t) = \mathbf{1}\{T(w) > t\}$ mean survival under baseline treatment choice w .

$$\Delta(t) = \mathbb{P}\{T(1) > t\} - \mathbb{P}\{T(0) > t\}$$

+0.10 means a 10 percentage-point survival advantage.

Specify the population, treatment choices and horizon.

This is a probability difference, not a hazard ratio or a relative increase.

The individual and the average

One person $\delta_i(t) = A_i^1(t) - A_i^0(t)$

Realized sample $\Delta_{\text{SATE}}(t) = \frac{1}{n} \sum_i \delta_i(t)$

Population $\Delta(t) = \mathbb{E}[\delta(t)]$

Similar profiles $\tau(t, x) = \mathbb{E}[\delta(t) \mid X = x]$

A conditional average for people like Hector is still an average.
It does not identify his missing potential outcome.

A fair comparison

Healthier people may be more likely to receive treatment and more likely to survive.

Adjustment needs measured common causes, treatment overlap and a well-defined treatment comparison.

Good prediction does not establish causal identification.

Exchangeability: $\{T(0), T(1)\} \perp W \mid X$

Overlap: $0 < \mathbb{P}(W = 1 \mid X = x) < 1$ for relevant profiles.

Censoring hides part of the assigned path

We record an event time or the end of observed follow-up:

$$Y = \min(T, C), \quad D = \mathbf{1}\{T \leq C\}.$$

If follow-up ends before t , survival at t may be unknown.

Censoring needs assumptions and adequate follow-up support.

It is separate from the missing counterfactual.

The effect can change over follow-up

Causal survival forests (CSF) estimate survival effects.

The website displays cohort averages at five horizons.

CAST fits a trajectory to those estimates:

$$\widehat{\Delta}_{\text{CAST}}(t) = \widehat{\beta}_0 + \widehat{\beta}_1 t + \widehat{\beta}_2 t^2.$$

The fitted line adds a model of the trajectory.

Additional drawing points add no observations. Keep the CSF points visible.

Read uncertainty at the right level

The same people contribute at several horizons,
so the estimates are correlated.

The demo carries that covariance into a
95% pointwise band around the fitted quadratic.

Pointwise is not simultaneous coverage of the whole curve.

The band does not capture all fit-selection or model-misspecification uncertainty.

A smooth fit can miss the trajectory

Frozen reversal scenario · 2,000 synthetic people

Measured confounding 1; hidden confounding 0. Differences in pp.

Month	Unadjusted	CSF	CAST	Oracle
12	13.49	6.51	6.39	7.51
36	32.35	18.09	10.34	16.38
60	28.56	11.42	9.96	9.15
84	14.51	−0.12	5.28	−1.59
108	6.81	−4.38	−3.72	−5.60

At which horizon is the CAST–oracle gap largest?

For whom, and for which question?

An effect may vary with baseline profile x :

$$\tau(t, x) = \mathbb{E}[A^1(t) - A^0(t) \mid X = x].$$

Name the outcome scale, horizon and population before describing heterogeneity.

These graphics show cohort averages, not subgroup estimates.

The effect of baseline treatment does not answer when to switch or stop it.

The class lab

Inspect the supplied 600-person synthetic cohort.

Run `class-lab.R`, or use the frozen results.

Compare five horizon estimates with the separate answer key.

Explain one discrepancy and one assumption the data cannot prove.

The small lab is a different sample from the website's 2,000-person export.

Record the target, horizon, units and uncertainty in your interpretation.

Return to the causal roadmap

Define the question and target population.

Connect the data through explicit assumptions.

Estimate, then interpret with uncertainty.

CSF and CAST belong inside that process.

Igor Shuryak's CAST repository: R implementation and methods

Hernán & Robins: causal foundations · Cui et al.: CSF

Yang et al.: CAST · Petersen & van der Laan: causal roadmap

Tao of RWD Navigator: a place to develop the study question